Lab 3

Section B01
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Experiment 1

Quantity	Measured	Expected (pre-lab / ideal)
$\langle v_d \rangle$	34.9 V	33.0 V
I_d	2.93 A	2.64 A
i_d min / max	−0.06 / 4.37 A	0 / 4.15 A
v_d max	51.7 V	51.9 V
V_{AB} (rms)	37.1 V	36.7 V
I_A (rms)	2.61 A	2.94 A
THD of i_g	2.1 %	0 %
THD of v_{ab}	3.0 %	0 %
PF	0.931	1.000

PF = P / ($V_{AB} \cdot I_A$) = 90 / (37.1 × 2.61) = 0.931. The recorded 45 W is one of the meter's two-phase readings, so total P ≈ 90 W.

Comparison with pre-lab 1 and PF.

The measured 34.9 V is 1.9 V higher than the pre-lab, but within the ±2.4 V scope accuracy. Pre-lab I_d is 0.29 A higher than the measured I_d, just outside ±0.24 A. $\langle v_d \rangle / I_d$ = 11.9 Ω suggests R is slightly below 12.5 Ω. For a resistive load i_g follows v_{ab}, so PF ≈ 1 is expected. The measured 0.931 is too low for a THD of only 2.1 %. The meter line currents read low at these small currents: I_A (2.61 A rms) is below I_d (2.93 A average), which is impossible since $|i_g| = i_d$. This error lowers the calculated PF.

Distortion of i_g. The transformer voltage is itself distorted (THD of v_{ab} = 3.0 %, flat-topped by other rectifier loads on the supply), and a resistor copies that shape into i_g. The two diode drops (≈0.85 V each) also keep i_g at zero briefly near each zero crossing, and L_c slightly delays each current transfer.

Experiment 2

Quantity	Measured	Expected (pre-lab / ideal)
$\langle v_d \rangle$	35.4 V	33.0 V
I_d	3.37 A	2.64 A
i_d min / max	0.39 / 4.85 A	0.43 / 4.11 A
120 Hz component of v_d	15.5 V pk	22.0 V pk
V_{AB} (rms)	37.4 V	36.7 V
I_A (rms)	2.37 A	2.91 A
THD of i_g	2.28 %	5.8 %
PF	0.934	0.991

Harmonics of i_g (I_h/I_1, %). Even orders measured as 0.

h	3	5	7	9	11	13
Constant i_d (1/h)	33.3	20.0	14.3	11.1	9.1	7.7
Measured	1.3	2.1	1.5	0.24	1.1	0.62

The recorded P (27 W) does not agree with $\langle v_d \rangle \cdot I_d \approx 119$ W, so the recorded PF of 0.934 is used.

Comparison with constant-i_d theory and Exp. 1. i_d stays positive (0.39 to 4.85 A) and i_g is i_d with its sign flipped each half cycle, but i_d is far from constant, so i_g is still close to a sine wave rather than the square wave from class. THD is 2.3 % against 48 % for constant i_d, and PF is 0.934 against 0.90. Compared with Exp. 1, the only visible change is that i_d no longer reaches zero; THD and PF are essentially unchanged. This agrees with pre-lab 2: $X = 2\pi \times 120 \times 5$ mH = 3.77 Ω is too small next to R = 12.5 Ω to hold i_d constant.

Inductor for 10 % ripple. The peak-to-peak ripple of v_R is about $2 \cdot V_{d2} \cdot R/|R + j\omega L|$. Setting this to $0.1 \cdot \langle v_d \rangle$ gives $L = (R/\omega) \cdot \sqrt{[(20 \cdot V_{d2}/\langle v_d \rangle)^2 - 1]} = (12.5/754) \cdot \sqrt{[(20 \times 15.5/35.4)^2 - 1]} = 0.14$ H. This is about 30 times the 5 mH inductor while carrying about 3 A dc, so it would be large, heavy and lossy, and not practical.

Experiment 3

Quantity	Measured	Expected (pre-lab / ideal)
$\langle v_d \rangle$	48.2 V	49.6 V
I_d	4.52 A	3.96 A
i_d min / max	3.90 / 4.91 A	3.60 / 4.15 A
v_d min / max	41.8 / 52.6 V	44.9 / 51.9 V
V_{aN} (rms)	21.2 V	21.2 V
V_{AB}, V_{BC}, V_{CA}	37.4, 37.5, 37.4 V	36.7 V
I_A, I_B, I_C	2.70, 2.75, 2.95 A	3.24 A
P	216.9 W	197 W
THD of i_{ga}	31.6 %	30.7 %
PF (meter)	0.939	0.956

PF cross-check: $P/(\sqrt{3} \cdot V_{LL} \cdot I_L) = 216.9/(\sqrt{3} \times 37.4 \times 2.80) = 1.19$, which is impossible and again shows the meter line currents read low. The meter PF (0.939) is used.

Comparison with pre-lab 3. Scaled to V_{LL} = 37.4 V, the expected values are $\langle v_d \rangle$ = 50.5 V, v_d max = $\sqrt{2} \times 37.4$ = 52.9 V and v_d min = 52.9 × cos 30° = 45.8 V. $\langle v_d \rangle$ is 2.3 V low, about the two diode drops (1.7 V) plus overlap, and within scope accuracy. v_d max matches. v_d min is 4 V low because [Meas] Minimum catches the commutation notches. I_d is 0.56 A high, beyond ±0.24 A; since $\langle v_d \rangle \cdot I_d$ = 218 W matches the meter P of 217 W, the load resistance is likely about 10.7 Ω rather than 12.5 Ω.

Shape of i_{ga}. Angles are from the rising zero crossing of v_{aN}. The top diode on the highest phase and the bottom diode on the lowest phase conduct:

Interval	Largest line voltage	Diodes conducting	i_{ga}
30° to 90°	v_{ab}	D1, D5	$+i_d$
90° to 150°	v_{ac}	D1, D6	$+i_d$
150° to 210°	v_{bc}	D2, D6	0
210° to 270°	v_{ba}	D2, D4	$-i_d$
270° to 330°	v_{ca}	D3, D4	$-i_d$
330° to 30°	v_{cb}	D3, D5	0

i_{ga} is a 120° positive block, a 60° gap, a 120° negative block and another 60° gap. Even with only R, v_d ripples only about 14 % at 360 Hz, so i_d is nearly flat and the tops of i_{ga} follow its small ripple.

Experiment 4

Quantity	Measured	Expected (pre-lab / ideal)
$\langle v_d \rangle$	48.2 V	49.6 V
I_d	4.50 A	3.96 A
i_d min / max	3.86 / 4.87 A	3.76 / 4.11 A
v_d min / max	41.8 / 52.6 V	44.9 / 51.9 V
V_{AB}, V_{BC}, V_{CA}	37.3, 37.3, 37.3 V	36.7 V
I_A, I_B, I_C	2.67, 2.72, 2.91 A	3.24 A
P	216.2 W	197 W
THD of i_{ga}	31.5 %	31 %
PF (meter)	0.94	0.955
Overlap time t_u	158 µs	285 µs
i_d at end of overlap	4.12 A	3.96 A

Harmonics of i_{ga} (I_h/I_1, %). Even orders measured as 0.01 % or less.

h	3	5	7	9	11	13
Constant i_d (1/h)	0	20.0	14.3	0	9.1	7.7
Measured	0	26.1	10.2	0.12	9.2	6.8

I_d, THD and PF. I_d = 4.50 A against 3.96 A predicted (3.95 A including the 0.14 V overlap drop); the excess matches Exp. 3 and has the same cause. THD is 31.5 % against 31 % and PF is 0.94 against 0.955, so the bridge behaves close to the ideal constant-i_d case. The PF difference is within the meter accuracy and supply distortion.

Harmonics compared with Exp. 2. Both currents have half-wave symmetry, so no even harmonics appear. i_g in Exp. 2 contains all odd orders, including 3 and 9, but they are small because i_g is nearly sinusoidal. i_{ga} in Exp. 4 contains only h = 6k ± 1 (5, 7, 11, 13): its 120° blocks cancel all triplen harmonics. Its harmonics are close to 1/h because i_d is nearly constant; the 5th is higher (26 %) because of the curved tops of i_d and the distorted supply voltage.

Overlap and L_c. $u = 360° \cdot f \cdot t_u = 360 \times 60 \times 158\ \mu s = 3.41°$, compared with 6.15° (285 µs) from pre-lab 3. $L_c = \sqrt{2} \cdot V_{LL} \cdot (1 - \cos u)/(2\omega \cdot i_d) = \sqrt{2} \times 37.3 \times (1 - \cos 3.41°)/(2 \times 377 \times 4.12) \approx$ 30 µH per line, about a third of the assumed 100 µH. Since $1 - \cos u$ grows as u^2, a 10 % error in placing the cursors on the slow start of the rise gives about 20 % error in L_c.

Comparison

Exp.	Case	$\langle v_d \rangle$ (V)	$\Delta i_{d,pp} / I_d$ (%)	THD (%)	PF
1	1-ph, R	34.9	151	2.1	0.931
2	1-ph, L-R	35.4	132	2.3	0.934
3	3-ph, R	48.2	22	31.6	0.939
4	3-ph, L-R	48.2	22	31.5	0.94

Ranking by PF. Measured: 4 ≈ 3 > 2 ≈ 1. From the shape of the source current the expected order is 1 > 2 > 3 ≈ 4: in Exp. 1 i_g is nearly sinusoidal and in phase with v_{ab} (PF ≈ 1), in Exp. 2 it is only slightly flattened (PF ≈ 0.99), and in Exp. 3 and 4 it is a 120° block with THD ≈ 31 % (PF ≤ 0.955). The measured Part A values do not follow this because, with THD near 2 %, they cannot physically be as low as 0.93; they are limited by the meter current error. The Part B values agree with theory.

Effect of the inductor, single vs three phase. From Exp. 1 to 2, i_d ripple fell from 151 % to 132 % and i_d min rose from 0 to 0.39 A, with no real change in THD (2.1 to 2.3 %). From Exp. 3 to 4 there was no measurable change in i_d ripple (22 % in both, near the probe noise and resolution at 1 A/div) or THD (31.6 to 31.5 %). In the single-phase bridge v_d falls to zero at 120 Hz, so the current follows it unless $\omega L \gg R$; 5 mH (3.8 Ω) is far too small. In the three-phase bridge v_d ripples only about 14 % at 360 Hz, so i_d is already nearly constant with R alone and the source current is already the ideal 120° block; the inductor has little left to change.

1 kW resistive load. Choose the three-phase bridge without a dc inductor. It loads all three transformer phases equally (1 kW / 0.955 ≈ 1.05 kVA, within the 1.2 kVA rating, while a single-phase line-to-line load would put about 580 VA on each of two 400 VA windings), gives a higher dc voltage (≈48 V, so lower current) and has low ripple without filtering. An inductor would add size, weight and copper loss for almost no gain. The trade-off is a PF of about 0.95 and a source current THD of about 31 %, compared with PF ≈ 1 for a single-phase bridge feeding R.